Glaze Search · how a search works (card G)

Usage mind map with the smart-search upgrade · NEW = added by the upgrade



docs/glaze_search_usage_mindmap.svg · step-by-step guide: docs/manuals/CG2_Glaze_Search_Step_by_Step_Guide.pdf

From data to evaluation · the whole machine-learning pipeline

Read left to right: what data goes in, how it is prepared, what is trained, and how it is measured. Every number is from the held-out test split.

Glaze Search · card G models

1 · DATA

ALL 7k Glazy recipe CSV
- 7,188 recipes · tags: cone, surface, transparency, atmosphere, subtype
- 26 oxide columns (+ rare ones)
- 4,859 free-text descriptions written by Glazy users

ALL ingredients CSV
- 49,263 rows: material, %, addition

PHOTO recipe photos
- 7,063 + 125 bonus photos → 12,309 segmented objects (flow 1)

ALL recipe bank
- 7,063 recipes · 7,052 with photo objects · 7,049 with a formula · 4,766 with a description

COLOUR MATCH colour-match model ZIP
- catalog 6,718 recipes · 256-d chemistry vectors · text head

2 · PREPROCESSING

TEXT cleaned tags → tag sentence
- Orton cones kept as text: 04 ≠ 4
- e.g. *cobalt blue glaze. Surface: glossy. Firing cone: 6.*

TEXT 8 noisy sentences per recipe
- typos (12 %), ^6 / c6 / cone6, shiny / matt, filler words, any word order

TEXT description = name + Glazy text

TEXT MiniLM-L6-v2 → 384-d
- mean pooling, ≤ 128 tokens (frozen)

PHOTO DINO → SAM 2.1 → CLIP → DINOv2
- 1024-d per object → area-weighted mean → L2

FORMULA oxide weight % → 26 fractions → √
- needs ≥ 3 oxides

COLOUR palette → CIELAB
- ColorThief top 3 colours + area · colour word = nearest of 98 shades (CIEDE2000)

ALL split by composition group
- 4,945 train · 1,059 validation · 1,059 test

3 · TRAINING

TEXT text reader: no training
- rules + regex + fuzzy matching (difflib ≥ 0.80)

SUGGESTIONS TF-IDF + logistic regression
- word 1–2 + char 3–5 grams · C = 4, balanced · from scratch on the train descriptions

JOINT MODEL encoders frozen - not fine-tuned
- 3 new heads: Linear → GELU → Dropout 0.1 → Linear, 512 → 256-d

JOINT MODEL symmetric InfoNCE, τ = 0.07
- text–photo · text–formula · photo–formula; same group is no negative

JOINT MODEL AdamW 1e-3 · wd 1e-4 · batch 256
- early stop on validation mean R@10 → best epoch 58 of 70 (about 2 min, GPU)

FAMILIES K-Means (k-means++, 10 restarts, Lloyd)
- k ∈ 8 / 10 / 12 / 15 / 20 by silhouette → 10 · HDBSCAN tried, not used

COLOUR MATCH colour match: not retrained
- used as delivered; parity with its notebook checked

4 · EVALUATION · test split

JOINT MODEL Recall@10 (right recipe in top 10)
- noisy text 74.2 % (MiniLM only 62.0 %)
- photo 45.3 % · formula 33.1 % · description 33.4 % · random 0.9 %

TEXT precision of read values
- colour 93.3 % · cone 99.2 % · surface 89.1 % · atmosphere 100.0 %
- 1,059 noisy sentences, 15 % typos

SUGGESTIONS accuracy vs majority
- surface 63.4 % vs 52.8 % · transparency 67.9 % vs 47.8 %
- atmosphere 92.7 % vs 87.9 % · colour family 36.7 % vs 37.4 % → not used

ALL P@10, colour + surface + cone all match
- text only 54.4 % · values only 55.3 % · **both 73.1 %**

FAMILIES silhouette · noise share
- K-Means 0.10 (weak) · HDBSCAN 2 clusters, 35 % noise

COLOUR MATCH parity + app tests
- same lists as the notebook, scores within 1e-5 · 38 tests · browser checks, 0 JS errors

parts: TEXT PHOTO FORMULA COLOUR JOINT MODEL SUGGESTIONS FAMILIES COLOUR MATCH ALL

Text → recognise → ceramic values

How the free text of **Describe it** becomes values, step by step, on a real example. Steps 1–6 are rules and fuzzy matching (no training); step 7 uses two trained models.

1 — Normalise

lower case, Unicode NFKC, grey → gray, 3+ repeated letters → 2

`shinyy cobolt blu glaze ^6 ox, kinda runny`

2 — Cone

regex: cone / c / ^ / Δ + Orton number or range (38 cones, 04 kept); low / mid / high fire → 06–04 / 5–6 / 9–10

`^6` → **cone 6**

3 — Amounts

regex: 26 oxide names + number (≥ 3 = an analysis); 22 colourant names with %

(none here; `8% red iron oxide` → Fe2O3 8 %)

4 — Keyword phrases

longest phrase first: 9 surfaces, 4 transparencies, 7 atmospheres, 10 effects and their spellings

`ox` → **Oxidation** · `runny` → **flowing**

5 — Typos

words left over → nearest keyword or colour word, difflib ratio ≥ 0.80 (≥ 0.85 for ≤ 4 letters)

`shinyy` → shiny → **Glossy** · `cobolt blu` → cobalt blue

6 — Colour → HEX

Color Lab V3 + 98 reference shades, character 2–4-gram TF-IDF; light / dark / -ish shift CIELAB; nearest model colour word by CIEDE2000

cobalt blue → **#0047AB** → *cobalt blue*

7 — Suggest + embed

TF-IDF + logistic regression: implied tags (shown at p ≥ 45 %) · the whole text → MiniLM → text head → 256-d

suggested **transparency opaque 50 %** · text vector for the joint model

What the page shows: colour **cobalt blue #0047AB** · cone **6** ← "^6" · surface **Glossy** ← "shinyy" · atmosphere **Oxidation** ← "ox" · effect **flowing** ← "runny" · typos fixed: shinyy→shiny, cobolt→cobalt, blu→blue · suggested transparency: opaque 50 %

Measured on 1,059 noisy sentences of test recipes (15 % typos)

value read	right when read	sentences with it
colour (ΔE00 ≤ 20 to the recipe's colour)	93.3 %	1,023
cone (Orton range overlaps)	99.2 %	667
surface	89.1 %	503
atmosphere	100.0 %	611

Why both rules and models

Written values (colour, cone, surface …) are read exactly by rules, so they can be shown and corrected. Everything else - unknown words like *tenmoku*, word order, style - is read by the trained text head. Together they find matching recipes far more often than either alone (top 10 matching colour + surface + cone: 73.1 % vs 54.4 % and 55.3 %).

What is pretrained, trained or fine-tuned · and which metric measures what

Models (no pretrained encoder is fine-tuned)

model	job	learnt from	here
Grounding DINO base	finds objects (text prompt → boxes)	pretrained	frozen · prompts and thresholds set
SAM 2.1 Hiera-Large	exact object outline	pretrained	frozen
OpenCLIP ViT-B/32 (LAION-2B)	is it ceramic?	pretrained	frozen · threshold 0.35
DINOv2 ViT-L/14	photo features (1024-d)	pretrained (self-supervised)	frozen
all-MiniLM-L6-v2	text features (384-d)	pretrained (sentence pairs)	frozen
joint heads (text, photo, formula → 256-d)	one space for all inputs	trained here on 7,063 recipes	**new layers trained** on top of the frozen encoders (InfoNCE) - the encoders are not fine-tuned
suggestion model	implied surface / transparency / atmosphere	trained here on 4,766 descriptions	trained from scratch
text reader	keywords + values from free text	vocabularies of the 7k data + Color Lab V3	rules, no learning
K-Means families	Explore formulations	fitted on the train split	k-means++ initialisation
colour-match model	colour + cone + surface search	trained before (6,718 recipes)	used as delivered, not retrained

Metrics

metric	measures	used for
Recall@K (K = 1, 5, 10)	share of queries whose own recipe (or a recipe of its composition group) is in the top K of 1,059	joint model: noisy text, description, photo, formula → recipe
median rank	the middle rank of the right recipe (1 = first)	same
precision of read values	when the reader gives a value, is it the recipe's value (colour: ΔE00 ≤ 20)	text reader
accuracy, macro-F1 vs always-the-most-common	is the suggestion model better than guessing	suggestion model
P@10, strict	share of the top 10 whose colour (ΔE00 ≤ 20), surface and cone all match the query recipe	the whole smart search (and without text / values)
validation mean Recall@10	early stopping: the epoch with the best mean of noisy text, photo and formula → recipe	training
silhouette · noise share	how well clusters separate; share of points HDBSCAN leaves unclustered	choosing K-Means k; rejecting HDBSCAN
score difference vs notebook	the app gives the notebook's numbers	colour match

Clustering: HDBSCAN (density-based, like DBSCAN) found 2 clusters and left 35 % of recipes as noise, so it cannot group recipes for Explore. K-Means with k-means++ initialisation always assigns every recipe; its silhouette is low (0.10), so its families are a practical way to show different recipes, not natural groups. They change which recipes Explore shows, never a score.

The process · steps 1–5 (a photo in card 1)

Model and settings of each step (pretrained models, used as they are)

1 Photo in

card 1 · app/server.py

- upload, paste, drop or link
- EXIF rotation
- longest side ≤ 2,048 px

2 Find objects

Grounding DINO base

- prompt: ceramic object · pottery piece · ceramic test tile · … · glazed pottery
- box threshold 0.24 · text threshold 0.25
- NMS IoU 0.60 · hands left out

3 Outline

SAM 2.1 Hiera-Large

- box → exact mask per object
- label map for hover and click

4 Ceramic?

OpenCLIP ViT-B/32

- p(ceramic) ≥ 0.35 → selected by default
- else the first object

5 Colour

pick · regions · palette

- pick: 13 × 13 px inside the mask, brightest / darkest 15 % out, median
- regions: SLIC 220 superpixels, merge ΔE76 < 11, min 1.2 %, median CIELAB
- names: nearest of 98 shades (CIEDE2000)

The process · steps 6–10 (every card G search)

Text reader, frozen encoders, the trained joint model, scoring and results

6 Read the text

rules + fuzzy matching

- see the text pipeline page
- typos: difflib ≥ 0.80
- suggestions shown at p ≥ 45 %

7 Encode

frozen encoders

- text: MiniLM-L6-v2, 384-d, ≤ 128 tokens
- photo: DINOv2 ViT-L/14, 1024-d
- formula: 26 oxides → √ fractions

8 Joint model

3 trained heads → 256-d

- 512 hidden, GELU, dropout 0.1
- InfoNCE τ 0.07
- best epoch 58

9 Score

parts you give, rescaled to 1

- text .35 · photo .35 · formula .25
- colour .30 · cone .10 · surface .10 · transp. .05 · atmos. .05
- colour: exp(−ΔE00 / 20) to the closest palette colour ≥ 10 %

10 Results

two lists, one ranking

- Best matches: top 10
- Explore: ≤ 2 per K-Means family (k = 10)
- Glazy photo m_ → l_ · View on Glazy

Smart search model · training and held-out test results

model 78afd7dfbe2c · trained 2026-11-04 00:02:20 · test split: 1,059 recipes the model never saw · random top 10 ≈ 0.9 %

Finding the right recipe

query → recipe	n	top 1	top 5	top 10	median rank
noisy sentence → recipe, all parts	1,059	31.7 %	61.9 %	**74.2 %**	3
noisy sentence → recipe text, pretrained MiniLM only	1,059	24.6 %	50.1 %	**62.0 %**	5
photo → recipe	1,058	14.3 %	34.9 %	**45.3 %**	14
tag text → recipe (photo + formula side)	1,059	9.9 %	28.6 %	**40.2 %**	16
real Glazy description → recipe	713	7.7 %	23.8 %	**33.4 %**	31
formula → recipe	1,058	6.6 %	23.7 %	**33.1 %**	26
noisy sentence → recipe (photo + formula side)	1,059	7.2 %	18.1 %	**28.2 %**	33

Training curve

Validation mean Recall@10 rose from 0.05 (epoch 1) to 0.36 (epoch 58); training stopped at epoch 70 after 12 epochs without gain.

Smart search on noisy sentences (300 test sentences)

scored with	top 10 matching colour + surface + cone
text model only	54.4 %
read values only	55.3 %
both = smart search	**73.1 %**

Suggestions (held-out descriptions)

tag	accuracy	always the most common	macro-F1
surface	63.4 %	52.8 % (glossy)	0.41
transparency	67.9 %	47.8 % (opaque)	0.64
atmosphere	92.7 %	87.9 % (oxidation)	0.61
family	36.7 %	37.4 % (gray)	0.27

Colour family is not better than always guessing "gray", so it is not shown in the app. Noisy sentences are generated from the recipes' tags; real user text can differ. Each PC that runs INSTALL_GLAZE_MODEL.bat trains its own model, so its numbers differ slightly (meta.json).

The whole system · three flows

Flow 1 builds the database once · Flow 2 searches objects by photo · Flow 3 is card G

